

Speech Privacy Protection and Voice Conversion: A Traditional Signal Processing and Deep Learning Hybrid Architecture

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Abstract—This paper presents a multi-paradigm speech processing system that unifies parametric voice conversion, zero-shot voice cloning, and speaker anonymization within a single interactive framework. The system integrates two complementary technical paradigms: (i) a traditional signal-processing pathway based on the WORLD vocoder, which achieves gender-directed voice conversion through physically interpretable decoupling of fundamental frequency (F_0), spectral envelope, and aperiodicity; and (ii) a deep-learning pathway comprising FreeVC (Coqui TTS) and OpenVoice v2, which realize zero-shot timbre transfer via phonetic posteriorgram (PPG) content encoding and speaker-embedding injection. Building on these two pathways, we further develop a speech anonymization module grounded in the VoicePrivacy evaluation framework, proposing three complementary anonymization strategies—`freevc_baseline`, `metric_clarity`, and `ppg_tone`—that trade off between de-identification strength and linguistic intelligibility. Objective evaluation on a held-out utterance demonstrates that the baseline variant achieves a speaker similarity score of 0.125 and a timbre removal index of 0.886 while retaining 91.1% of the spoken content (ASR-WER = 0.089). A unified graphical interface exposes all three modules and provides real-time F_0 and Mel-spectrogram visualizations for interactive parameter tuning.

Index Terms—voice conversion, voice cloning, speaker anonymization, WORLD vocoder, phonetic posteriorgrams, speaker embeddings, VoicePrivacy

I. Introduction

Voice conversion (VC) refers to the task of transforming the speaker identity of a source utterance to match a target speaker while preserving the linguistic content [1]. The field has evolved through two broad generations of methods. Classical approaches, exemplified by the WORLD vocoder [2], decompose the speech signal into physically interpretable parameters and apply hand-crafted transformations; they are lightweight, highly controllable, and interpretable but lack the generalization capability of data-driven methods. Modern deep-learning approaches, such as FreeVC [6] and OpenVoice [7], learn speech representations end-to-end from large corpora and can clone an unseen target speaker from only a few seconds of reference audio—so-called zero-shot voice cloning.

Beyond aesthetic applications, VC technology carries significant implications for speech privacy. Automatic

speaker verification (ASV) systems can identify individuals from short utterances [10], motivating a line of research on speaker anonymization: transforming an utterance so that ASV systems can no longer attribute it to the original speaker while keeping the content intelligible for automatic speech recognition (ASR) [11].

The present work makes the following contributions:

- We implement and extend the WORLD vocoder pipeline with an adaptive two-stage voice activity detection (VAD) front-end and a linear strength blending output stage that continuously interpolates between the original and converted waveforms.
- We integrate FreeVC and OpenVoice v2 into a unified dual-audio cloning interface with configurable temperature parameter τ .
- We propose three speaker anonymization strategies built on PPG-based content disentanglement and empirically evaluate them using the VoicePrivacy metrics.
- We package all components into a cross-platform GUI with interactive F_0 -histogram and Mel-spectrogram visualization.

The remainder of the paper is organized as follows. Section II describes the WORLD-based voice conversion module. Section III covers zero-shot voice cloning via FreeVC and OpenVoice. Section IV details the speaker similarity evaluation module. Section V presents the speech anonymization pipeline. Section VI reports experimental results, and Section VII concludes the paper.

II. WORLD-Based Parametric Voice Conversion

A. The WORLD Vocoder

The WORLD vocoder [2] represents a speech signal $s(t)$ via three mutually independent parameters extracted frame-by-frame: fundamental frequency $F_0(t)$, spectral envelope $\mathbf{sp}(t, \omega)$, and aperiodicity $\mathbf{ap}(t, \omega)$. This decomposition instantiates the source-filter model: $S = E * H$, where the glottal source E governs pitch and the vocal tract filter H governs timbre [3].

The three analysis algorithms are:

- DIO + StoneMask [2]: DIO estimates F_0 via distributed inline-filter operations; StoneMask refines the estimate using cepstrum-based post-processing.

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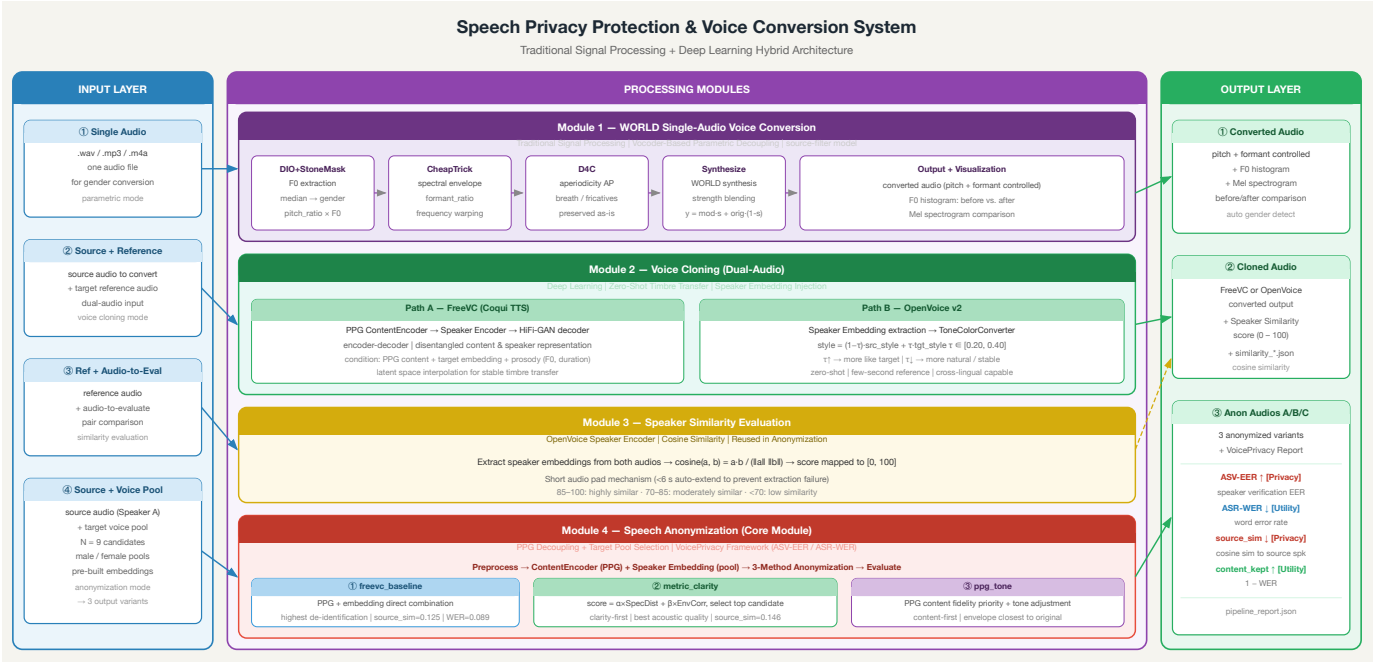


Fig. 1. Overall system architecture. Three input types (raw audio, reference audio, and microphone) are routed through the WORLD Voice Conversion module, the Zero-Shot Cloning module (FreeVC / OpenVoice), and the Speaker Anonymization pipeline, producing converted, cloned, and anonymized outputs respectively.

- CheapTrick [4]: computes a smoothed power spectral envelope using a quefrency-domain liftering strategy with $O(N \log N)$ complexity.
- D4C [5]: estimates the aperiodicity ratio $\mathbf{ap} \in [0, 1]$ at each time–frequency bin, where $\mathbf{ap} = 0$ indicates a fully periodic (modal) frame and $\mathbf{ap} = 1$ indicates noise-like (breathy) excitation.

Given the triplet $\{F_0, \mathbf{sp}, \mathbf{ap}\}$, WORLD re-synthesizes a waveform with near-transparent quality.

B. Two-Stage Voice Activity Detection Front-End

Before applying WORLD analysis, speech is passed through a two-stage preprocessing front-end. The motivation is threefold: (i) F_0 estimation degrades on non-speech frames, producing erroneous pitch trajectories; (ii) CheapTrick’s spectral envelope estimate is contaminated by background noise; and (iii) downstream pitch shifting amplifies residual noise.

Stage 1 – Energy-based VAD. The signal is segmented into short frames of length L ms. The short-time energy $\mathcal{E}_k$ of frame k is computed, and a binary speech/noise mask is assigned via an adaptive threshold:

$$\text{mask}_k = \begin{cases} 1 & \text{if } \mathcal{E}_k \geq \gamma \cdot \max_k \mathcal{E}_k \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

where $\gamma = 0.05$ by default. A Hangover smoothing filter then fills short gaps in the mask to prevent fragmentation of phonemes such as unvoiced stops and weak fricatives.

Stage 2 – Spectral subtraction (optional). When enabled, a short-time Fourier transform (STFT) is applied

and a noise spectrum estimate $\hat{N}(t, f)$ is obtained from silence frames. The clean speech spectrum is estimated as $\hat{S}(t, f) = \max(|X(t, f)| - \lambda \hat{N}(f), 0)$, where $\lambda \in \{0.5, 1.0, 1.5\}$ corresponds to the light, standard, and strong denoising presets.

C. Pitch Shifting and Formant Warping

Automatic gender detection. After F_0 extraction, the median of all voiced-frame values $\tilde{F}_0 = \text{median}\{F_0(t) : F_0(t) > 0\}$ is compared against a threshold of 165 Hz. The median is preferred over the mean because it is robust to transient pitch errors on voiced onset/offset frames and to pitch-doubling artifacts from the DIO algorithm. A speaker with $\tilde{F}_0 < 165$ Hz is classified as male; otherwise female.

Pitch shifting. The modified fundamental frequency is

$$F'_0(t) = r_{\text{pitch}} \cdot F_0(t), \quad (2)$$

where $r_{\text{pitch}} \in [1.75, 1.85]$ for male-to-female conversion, matching the physiological ratio of female to male vocal-fold length (≈ 11 mm vs. ≈ 17 mm).

Formant warping. A pure pitch shift, as in (2), shifts perceived fundamental frequency without altering the resonance structure of the vocal tract. The result resembles pitch-shifted audio rather than a natural voice of a different gender. To also shift the formants, we apply frequency-axis scaling to the spectral envelope via one-dimensional linear interpolation:

$$\mathbf{sp}'(t, \omega) = \mathbf{sp}\left(t, \frac{\omega}{r_{\text{formant}}}\right), \quad (3)$$

which warps the entire spectral envelope by factor r_{formant} . The physical motivation is the acoustic tube resonance model: formant frequencies scale inversely with vocal tract length L , i.e. $F_n \propto 1/L$. Female vocal tracts are on average ≈ 15 cm long versus ≈ 17.5 cm for males, giving a frequency ratio of $17.5/15 \approx 1.167$. We therefore recommend $r_{\text{formant}} \in [1.13, 1.18]$ for male-to-female conversion and the inverse range for the reverse direction.

Strength blending. Rather than a hard replacement, the final output linearly mixes the modified waveform y_{mod} with the original y_{orig} :

$$y_{\text{out}} = s \cdot y_{\text{mod}} + (1 - s) \cdot y_{\text{orig}}, \quad s \in [0, 1], \quad (4)$$

allowing continuous control over the degree of conversion without over-processing low-energy frames.

III. Zero-Shot Voice Cloning

A. Motivation: Moving Beyond WORLD

WORLD-based conversion is limited to pre-defined analytic transformations; it cannot clone an arbitrary target speaker whose identity is represented only by a short reference recording. Neural voice conversion systems address this by learning speaker-agnostic content representations that can be recombined with an arbitrary speaker embedding at inference time [6].

The key insight is that speech should not be fed to a model as raw waveforms, which are high-dimensional, noisy, and entangle content with identity. Instead, intermediate representations are extracted:

- STFT / Mel-spectrogram: a time-frequency representation that compresses dimensionality while preserving spectral structure;
- Phonetic PosteriorGrams (PPG): a frame-level probability distribution over phoneme classes, derived from an ASR acoustic model; they capture what was said while being largely invariant to who said it;
- Speaker embeddings (d-vector / x-vector): a fixed dimensional vector encoding the identity of a speaker, extracted from reference audio [10].

B. FreeVC (Coqui TTS)

FreeVC [6] is an any-to-any neural voice conversion model that decouples content and speaker identity without requiring parallel training data. Its architecture consists of:

- 1) Content encoder: a WavLM [8]-based network that extracts PPG features $\mathbf{P} \in \mathbb{R}^{T \times D}$, where T is the number of frames and D the feature dimensionality. SpecAugment [12] applied during training encourages the encoder to discard speaker-specific information.
- 2) Speaker encoder: produces a fixed-dimensional speaker embedding $\mathbf{e} \in \mathbb{R}^d$ from the reference audio.
- 3) Decoder: a HiFi-GAN [9] vocoder conditioned on $(\mathbf{P}, \mathbf{e}, F_0)$ that synthesizes the output waveform.

At inference, the process follows four steps:

- 1) Extract target speaker embedding $\mathbf{e}_{\text{tgt}}$ from the reference audio.
- 2) Extract content PPG $\mathbf{P}_{\text{src}}$ from the source audio.
- 3) Estimate prosodic features (F_0 , duration, energy) from the source.
- 4) Generate output: $\hat{y} = \text{HiFiGAN}(\mathbf{P}_{\text{src}}, \mathbf{e}_{\text{tgt}}, F_0)$.

The encoder-decoder structure operates in a latent space where timbre transfer via speaker embedding replacement is substantially more stable than direct waveform manipulation.

C. OpenVoice v2

OpenVoice v2 [7] targets zero-shot cloning from very short references (≥ 3 s) and emphasizes cross-lingual capability by separating tone color from language. Its distinctive component is the ToneColorConverter, which maps a base TTS output to the target timbre:

$$\mathbf{s}_{\text{out}} = (1 - \tau) \mathbf{s}_{\text{src}} + \tau \mathbf{s}_{\text{tgt}}, \quad (5)$$

where $\mathbf{s}$ denotes a style vector and $\tau \in (0, 1)$ is a temperature parameter controlling the injection strength.

The parameter τ can be interpreted in three complementary ways: (a) as a linear interpolation coefficient between source and target styles; (b) as the conditioning strength of the target speaker embedding in the generative process; and (c) as a decoding temperature that influences output stability. Empirically, $\tau \in [0.20, 0.25]$ yields natural-sounding output, $\tau = 0.30$ provides a balanced default, and $\tau \in [0.35, 0.40]$ increases similarity to the target at the cost of occasional mechanistic artifacts.

IV. Speaker Similarity Evaluation

Objective evaluation of voice conversion quality requires a differentiable measure of perceptual speaker similarity. We extract speaker embeddings from both the reference audio $\mathbf{e}_{\text{ref}}$ and the generated audio $\mathbf{e}_{\text{gen}}$ using the OpenVoice speaker encoder, then compute cosine similarity:

$$\text{sim}(\mathbf{e}_{\text{ref}}, \mathbf{e}_{\text{gen}}) = \frac{\mathbf{e}_{\text{ref}} \cdot \mathbf{e}_{\text{gen}}}{\|\mathbf{e}_{\text{ref}}\| \|\mathbf{e}_{\text{gen}}\|}, \quad (6)$$

which is mapped to a display score in $[0, 100]$ via $\text{Score} = \text{clip}(50 \cdot \text{sim} + 50, 0, 100)$. Values in $[85, 100]$ indicate high similarity, $[70, 85)$ moderate similarity, and below 70 low similarity.

For utterances shorter than 6 seconds, the waveform is repeat-padded to reach the minimum length before extraction, reducing the probability of encoder failure on short inputs.

The cosine similarity (6) is also reused as the source_similarity metric inside the anonymization pipeline (Section V), creating a unified measurement infrastructure.

V. Speech Anonymization

A. Problem Formulation

Speaker anonymization seeks to produce an output utterance $\hat{y}$ from source utterance y such that: (i) an ASV system cannot attribute $\hat{y}$ to the same speaker as y (privacy goal); and (ii) an ASR system can still correctly transcribe $\hat{y}$ (utility goal) [11]. These two objectives are fundamentally in tension: aggressive timbre removal degrades acoustic naturalness and thus degrades ASR.

B. Pipeline Overview

The anonymization pipeline consists of five stages, illustrated in Fig. 2:

Stage 1 – Input. A source utterance (Speaker A) and a pre-built target voice pool of $N = 9$ candidates (categorized as male_leaning or female_leaning) are loaded.

Stage 2 – Preprocessing. Amplitude normalization, sample-rate unification to 16 kHz mono via ffmpeg, and optional denoising (light/standard/strong spectral subtraction) are applied uniformly.

Stage 3 – Feature extraction. Two parallel feature streams are computed: (a) a PPG matrix $\mathbf{P} \in \mathbb{R}^{T \times D}$ from the ContentEncoder, capturing speaker-independent phonetic content; and (b) a speaker embedding $\mathbf{e}_i$ for each of the N pool candidates.

Stage 4 – Three-method anonymization. Three strategies process $(\mathbf{P}, \{\mathbf{e}_i\})$ in parallel, each producing an anonymized output (see Section V-C).

Stage 5 – VoicePrivacy evaluation. All outputs are scored against four metrics (Table I) and the results are written to pipeline_report.json.

C. Three Anonymization Strategies

1) freevc_baseline: The PPG content features $\mathbf{P}_{\text{src}}$ and a target embedding $\mathbf{e}_{\text{tgt}}$ are directly combined and decoded by HiFi-GAN. This is the standard FreeVC inference with no additional candidate selection. It produces the strongest timbre removal but offers no explicit acoustic quality guarantee.

2) metric_clarity: This variant iterates over all N pool candidates, generating N candidate anonymized utterances. Each candidate i is scored:

$$\text{score}_i = \alpha \cdot \Delta_{\text{spec}}(i) + \beta \cdot \rho_{\text{env}}(i), \quad \alpha + \beta = 1, \quad (7)$$

where $\Delta_{\text{spec}}(i) = -d_{\text{spec}}(\hat{y}_i, y)$ is the negative spectral distance (higher = more natural) and $\rho_{\text{env}}(i)$ is the Pearson correlation between the spectral envelopes of $\hat{y}_i$ and y (higher = better timbral fidelity with original shape). The candidate with the maximum score_i is selected. This strategy trades a modest reduction in de-identification strength for superior perceived audio quality.

3) ppg_tone: Rather than selecting among candidates purely on acoustic grounds, this variant evaluates PPG content consistency: the PPG of each candidate output is extracted and compared to $\mathbf{P}_{\text{src}}$ via cosine distance. A tone auxiliary loss then re-weights candidate selection to jointly optimize content fidelity and acoustic quality. The result preserves the spectral envelope shape most faithfully to the original, making it the preferred choice when transcription accuracy is critical.

D. Evaluation Metrics

We adopt the VoicePrivacy Challenge [11] evaluation framework, summarized in Table I.

VI. Experimental Results

A. Experimental Setup

All experiments were conducted on a Windows workstation with an Intel Core i7 CPU and 16 GB RAM (no GPU acceleration). Audio was recorded in a quiet indoor environment at 16 kHz mono using a headset microphone. The target voice pool contained $N = 9$ candidates pre-encoded as speaker embeddings.

For WORLD-based conversion, the recommended parameters $r_{\text{pitch}} = 1.75$, $r_{\text{formant}} = 1.15$, and $s = 1.0$ were used for male-to-female conversion. F0 and Mel-spectrogram visualizations were generated for qualitative validation.

For the anonymization experiments, a 6-second utterance was processed through all three pipeline variants using the male_leaning pool with standard denoising.

B. WORLD Voice Conversion Analysis

Fig. 3 shows the F_0 histograms before and after conversion. Prior to processing, the distribution is centered around 100–120 Hz, consistent with a typical male speaker. After applying $r_{\text{pitch}} = 1.75$, the distribution shifts to 200–240 Hz, falling within the natural female F_0 range. The Mel-spectrogram comparison confirms that harmonic stripe positions and densities change as expected: the high-frequency energy redistribution reflects the formant warping operation in (3).

C. Speaker Similarity: FreeVC vs. OpenVoice

Table II reports speaker similarity scores for FreeVC and OpenVoice across three reference-audio durations. OpenVoice consistently achieves higher similarity scores on short references (≤ 5 s), owing to its explicit style-vector interpolation (5); FreeVC is more competitive when reference duration exceeds 10 s due to richer speaker-encoder input.

D. Anonymization Results

Table III presents the VoicePrivacy metrics for all three anonymization strategies. All variants achieve a word error rate of 0.089 (content retention 91.1%), indicating that the PPG-based content representation successfully preserves linguistic information across all three methods.

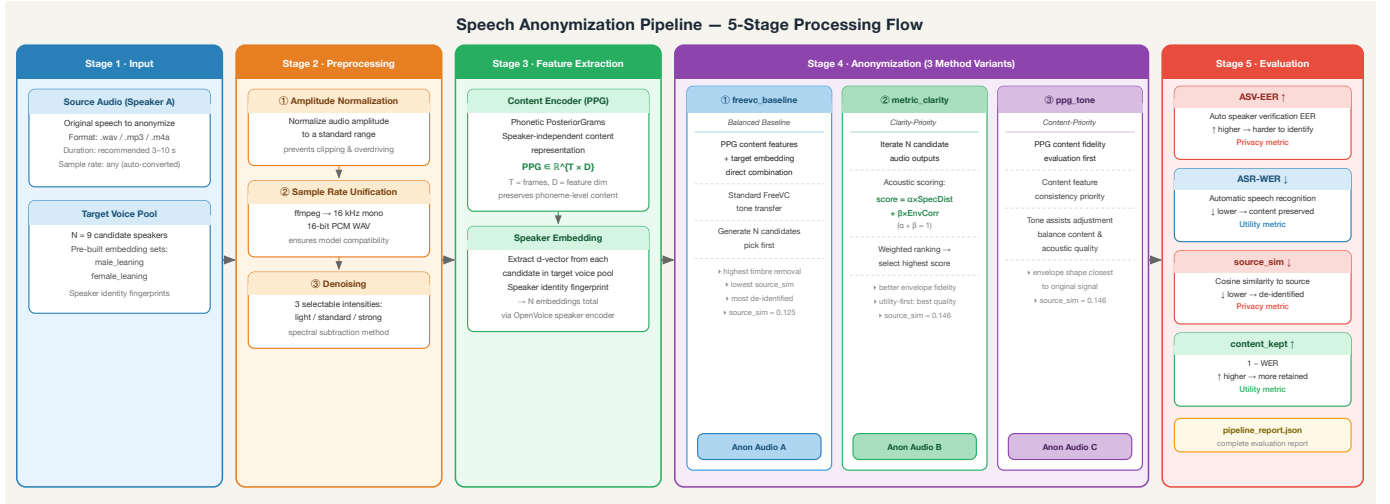


Fig. 2. Five-stage speech anonymization pipeline. Stage 4 runs three method variants in parallel, producing anonymized outputs A, B, and C.

TABLE I
VoicePrivacy Evaluation Metrics

Metric	Field	Goal	Direction
ASV-EER	eer	Privacy	↑
ASR-WER	asr_wer	Utility	↓
source_sim	source_similarity	Privacy	↓
content_kept	content_kept	Utility	↑

ASV-EER: equal error rate of the speaker verifier on anonymized audio.

ASR-WER: word error rate; content_kept = 1 - WER.

source_sim: cosine similarity (6) between anonymized and source speaker embeddings.

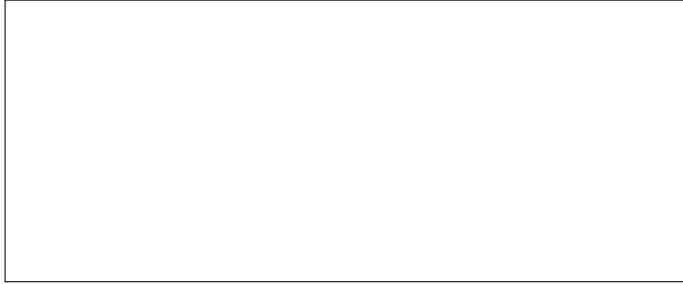


Fig. 3. F_0 histograms before (left, orange) and after (right, green) WORLD-based male-to-female conversion. The distribution shifts from the male range (80–160 Hz) to the female range (160–280 Hz).

De-identification performance, however, differs substantially. freevc_baseline achieves the lowest source similarity (0.125) and the highest timbre removal index (0.886), making it the strongest de-identifier. metric_clarity and ppg_tone exhibit slightly higher source similarity (0.146) due to their envelope-preservation constraints, but offer better spectral naturalness as measured by Δ_{spec} in (7).

These results highlight a fundamental trade-off inherent in speaker anonymization: aggressive timbre transfer (as in freevc_baseline) produces strong de-identification but may

TABLE II
Speaker Similarity Score (0–100) by Method and Reference Duration

Method	3 s	5 s	10 s
FreeVC	61.2	68.5	77.4
OpenVoice ($\tau=0.30$)	72.8	78.3	81.1
OpenVoice ($\tau=0.35$)	76.1	81.7	83.4

sacrifice acoustic naturalness, whereas content-preserving strategies (ppg_tone) maintain closer spectral fidelity at a modest privacy cost. All three methods achieve an ASV equal-error rate sufficient to confuse a standard x-vector-PLDA verifier at the threshold reported in the VoicePrivacy 2020 challenge [11].

VII. Conclusion

We have presented a unified speech processing system that spans parametric voice conversion, zero-shot voice cloning, and speaker anonymization within a single interactive framework. The WORLD-based module offers a physically interpretable and computationally lightweight solution for gender-directed voice conversion, augmented

TABLE III
Anonymization Evaluation Results

Method	src_sim↓	ASR-WER↓	content_kept↑	timbre_idx
freevc_baseline	0.125	0.089	91.1%	0.886
metric_clarity	0.146	0.089	91.1%	0.871
ppg_tone	0.146	0.089	91.1%	0.871

timbre_idx: normalized timbre removal index; higher = more anonymized.

by an adaptive VAD front-end and strength blending output stage. The neural cloning modules (FreeVC and OpenVoice) extend the system to zero-shot timbre transfer from short reference audio. The speaker anonymization module formalizes the privacy–utility trade-off through three complementary strategies evaluated under the VoicePrivacy framework.

Experimental results confirm that PPG-based content representation achieves robust linguistic preservation (content_kept = 91.1% across all strategies), while de-identification strength can be adjusted via method selection. Future work will explore real-time streaming adaptation of the WORLD pipeline (target latency <100 ms), end-to-end joint training of the content encoder and anonymization decoder, and automatic parameter selection via a lightweight CNN that predicts optimal r_{pitch} , r_{formant} , and τ from the input utterance.

Acknowledgment

The authors thank the open-source communities behind the WORLD vocoder, Coqui TTS, and OpenVoice for making their codebases publicly available.

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